

does it increase security, it also makes sharing of 3D resources easier.

As you might know, under Vista, each window is a 3D surface. Thus a graphics card under Vista is always called for, whether or not you're playing a game. Moreover, with the advent of high-definition video content, more and more 2D acceleration tasks will be offloaded to the GPU. The per-pixel shading capabilities of a GPU will be utilised to present everything from fancy transparent windows to smooth 720p video playback. Similarly, animations such as windows flipping to the foreground or cascading behind each other will be accelerated by the GPU. DX10 will thus expose the hardware to more and more applications, much like how the CPU evolved.

Having said that, the user-interface of Vista does not require DX10—the Aero Glass interface, as it's called, runs under a variant of DirectX 9. This was done to ascertain that development of Vista is not slowed down by the development of DX10 (both were concurrent developments). Elements of DX9 used in Vista are also seen under DX10, and in the future, the entire interface might move to a DX10 environment—once DX10 reaches maturity and is deemed stable enough.

Windows Vista will thus ship with both DirectX 9.0 and 10. DirectX 10 is currently slated to be a feature exclusive to Windows Vista.

The Ever-Distant Horizon

It's time for déjà vu yet again. We have been promised graphics nirvana before—virtually every piece of graphics hardware ever sold has been accompanied by shouts of “we have arrived.” Reality, however, walks a generation or two behind the latest and the greatest. With DX10, we should not expect things to change overnight. DX10 promises games that are more visually complex, offering a more compelling interaction and narrative. That promise will not be fulfilled with the release of Vista, but will see fruition a couple of years thence. And by then we would have forgotten all about DX10—and be talking about the next iteration.

By all indications, DX10 video cards will hit the half-a-billion transistor count, and will be so power-hungry that they will require dedicated power supplies. On the positive side, DX10 video cards will be virtually identical in the features that they offer. A DX10 card will have little to differentiate itself from a similar offering: this has been done largely on the behest of game developers who are tired of programming for esoteric feature sets (remember ATI's TruForm?). This is a huge plus for gamers as well. For example, with DX9 class cards, one had to worry whether the card supported Shader Model 2.0 or 3.0; concerns of that nature need no longer send us scurrying to the nearest search engine, or to our friendly-neighbourhood 3D guru.

DirectX 10 might not bring about a real-time *Toy Story* to our monitors; it will certainly not bring about graphics comparable to the real world, but it is a vital step towards those horizons. ■

readersletters@thinkdigit.com



Specialize in Networking & Security

@
Best IT Infrastructure Labs in India

- ♦ **Lab:** 60 Cisco Routers (3600, 2600, 2500, 1700, 800 Series), 15 Catalyst Switches (5000, 3750, 3550, 3000, 2950, 2900, 1900 Series), ISDN / Frame-Relay / X.25 WAN setup, Ethernet / Wireless / Token-Ring LAN Setup, CiscoPixSIE/ 501 Firewalls, Nokia IP 330 VPN Appliance, Cisco 3005 VPN Concentrator, Cisco 4210 IDS Sensor, Sonicwall, Watchguard Firewall.
- ♦ **Students Placed @** Wipro, C&W, TCS, Microland, HCL Technologies, HP, VSNL, IBM, I.T.I, Sify, Intel, Cisco.

First Time In India

CCIE Program for Fresh Graduates

Cisco Certified Internetwork Expert (CCIE) Training

CCIE: Security Track

CCIE: Routing & Switching Track

**Guaranteed Job in USA
with H1-B Visa**

Courses Offered:-

Networking : CCIE, CCIP, CCNP, CCNA

Wireless Networking : CWSP, CWNA

Security : CCIE, CCSP, CISSP/CCSA/CCSE

System Administration : Microsoft MCSE 2003

LEAD INSTRUCTOR :

Mr. AJAY PANDEY, B.E, MS (USA)
CCIE # 14792, (Security / R & S)
CISSP # 43675, CCSE, MCSE, ASAP.
Most qualified Network/Security instructor in India.
Ex-DeloitteConsulting & Lucent Technologies., USA

CONGRATULATIONS

We Congrate our Lead Instructor
MR. AJAY PANDEY on becoming
an double CCI - Security / R & S



We Accept

Top Gun Network Technologies Pvt. Ltd.

Corporate Office :

650/12, Bee Em Avenue, Dr. Rajkumar Road,

Rajajinagar Bangalore -560 010

Tel : 080 - 2352 8500/ 5761 6600

E-mail: info@topguntechnologies.com
www.topguntechnologies.com

**If You Find A Better Insititute To
Learn Any Of These Courses, Join It**